Yuechen Guo

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Education

University of Washington Sep 2024-Present
Master of Science in Electrical Engineering *Seattle, WA*

- **Relevant Coursework:** Data Structures and Algorithms, Database Systems, System and Software Tools (Linux), Mobile Applications Development (Android) , Software Engineering (Java), Computer Systems Security
- **GPA:** 3.9/4.0

Sun Yat-sen University Sep 2020-Jun 2024
Bachelor of Engineering in Electronic Information Science and Technology *Guangzhou, China*

Professional Experience

Chatbook AI Seattle, WA | Jul 2025 – Sep 2025

- Developed backend services as part of a 5-person team, building an enterprise **REST API** for a service appointment platform; implemented **AWS** configurations, **CORS** policies, and unit tests, while closely collaborating with frontend engineers to refine request/response validation and ensure reliable transactions.
- Implemented payment features with **Django**, integrating HelcimPay.js/Helcim API for secure checkout and card tokenization to ensure **PCI DSS** compliance, while applying security methods such as **hash validation** to maintain **payment integrity**.
- Built and launched an **AWS**-based asynchronous payment pipeline using **EventBridge, Lambda, and SQS**, designed to handle transaction outcomes from third-party processors (Helcim/Stripe), achieving 1,000+ daily transactions at **99.5%** success with idempotency, DLQs, and automated retries.
- Built an AI voice agent with **Twilio** integration, leveraging **OpenAI Realtime API and Agents SDK** to enable low-latency conversations **less than 500ms** with natural barge-in and function calling, automating 5+ call workflows including booking, rescheduling, and record updates.

Projects

VoiceShield: Audio Deepfake & Duress Detection System | *Capstone project with BECU*

* Built a **FastAPI** backend on **AWS EC2** with a **JavaScript** frontend for real-time audio upload and deepfake/duress detection visualization; optimized concurrency with **async I/O** and **thread pooling**, reducing latency by **50%**.
* Developed a duress detection pipeline with OpenSMILE feature extraction, SHAP+RFE feature selection, and an **LSTM–Random Forest ensemble**, achieving **93%** accuracy on real-world call data.

Personal Website | *Next.js + Spring Boot + PostgreSQL*

* Developed a full-stack personal website with **RESTful backend services**, integrated **CI/CD pipelines** via **GitHub Actions** and **Docker**, and implemented **unit tests** achieving **70%** code coverage with focus on critical path.
* Optimized article query logic by introducing JOIN FETCH to eliminate the **N+1 problem**, adding **composite indexes** for efficient filtering, and integrating high-performance **in-memory caching**; reduced average article query latency from **280 ms** to **20 ms** and scaled to hundreds of concurrent requests with stable performance.
* Implemented a scalable like/vote feature leveraging **Redis** and **Lua scripts** to ensure atomic operations, prevent race conditions under **high concurrency**, and reduce database load through scheduled asynchronous persistence.

Campus News Information Platform | *Spring Boot + Spring Cloud + MyBatisPlus*

* Built **microservice architecture** with modules for article publishing, removal, asynchronous review, and user interactions; managed service configuration via **Nacos** and routed requests through **Spring Cloud Gateway.**
* Implemented **JWT-based** authentication in the gateway microservice using **Spring Security**, storing user information in **ThreadLocal** within microservice interceptors to ensure unified validation and thread isolation.

Skills

- **Programming Languages**: Java, Python, C++, JavaScript, TypeScript, SQL, HTML/CSS, MATLAB
- **Frameworks & Tools**: Spring Boot, Spring Cloud, Django, Node.js, Next.js, React, Tailwind CSS, Bootstrap, MySQL, PostgreSQL, MongoDB, Redis, Git, Shell, Docker, AWS, JUnit, JMeter, Postman, Cursor IDE
- **Core Competencies**: Data Structures and Algorithms, Object-Oriented Programming (OOP), System Design
- **Languages**: English (Fluent), Chinese (Native), Japanese (Beginner)